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Contribution of molecular studies to botanical epidemiology and disease modelling: grapevine downy mildew as a case-study

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Abstract After being accidentally introduced from the USA at the end of the 19th century, downy mildew caused by *Plasmopara viticola* (Berk. *et* Curt.) Berlese *et* De Toni became one of the most damaging diseases affecting *Vitis vinifera* in Europe. Downy mildew causes both direct and indirect losses and can lead to severe reduction of yield. Our understanding of the life cycle and epidemiology of *P. viticola* has been recently altered by molecular studies that revealed that the overwintering inoculum (i.e., the oospores) does more than initiate disease, as was previously thought. A mechanistic model was developed for predicting the entire chain of processes leading to primary infections, and this primary infection model was linked to other models of secondary infection cycles. The model for primary infections defines the length of the primary inoculum season and a seasonal oospore dose consisting of several cohorts of oospores that progressively mature. The model was evaluated by means of Bayesian analysis in both Italy and eastern Canada, and showed high sensitivity, specificity, and accuracy both for potted plants and vineyards. Fungicide applications are necessary to control downy mildew because preventive agronomic practices are not very effective, including host resistance. The use of warning systems based on weather-driven models leads to a reduction in the use and cost of chemicals and a reduction in their environmental impact.

Keywords *Plasmopara viticola* · Life cycle · Oospores · Dynamic modelling · Fungicide scheduling

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Introduction

Plasmopara viticola (Berk *et* Curt.) Berlese *et* de Toni is the causal agent of grapevine downy mildew, an important disease in all grape-growing areas with frequent rains (Lafon and Clerjeau 1988). *Plasmopara viticola* was accidentally introduced in Europe at the end of the 19th century, most probably on rootstocks resistant to grape phylloxera (*Phylloxera vastatrix*) that were imported from North America and grafted with the European varieties (Viennot-Bourgin 1949). After the first report of the presence of downy mildew in vineyards in the late 1870s, the disease rapidly spread throughout Europe and North Africa (Fig. 1).

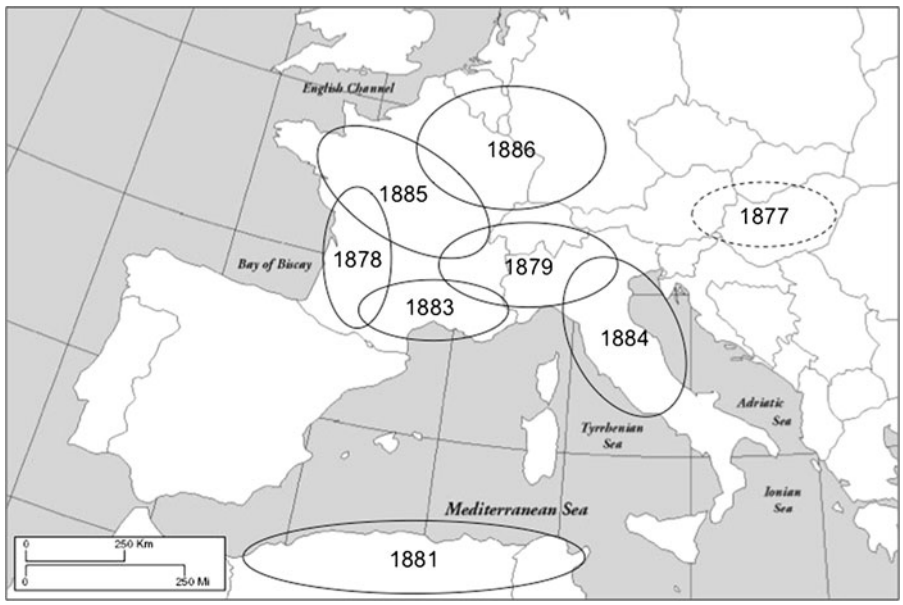


Fig. 1 Date and area of first report of *Plasmopara viticola* across Europe and North Africa. The dotted line indicates an unconfirmed appearance (data from Galet, 1977)

Because downy mildew caused and continues to cause substantial economic loss in Europe, hundreds of research projects aimed at understanding the pathogen and managing the disease have been conducted since the late 19th century. The final classification was proposed by Berlese and de Toni in 1888. New taxonomic tools and new molecular tools in particular have demonstrated that *P. viticola* belongs to the kingdom of *Protista*, subkingdom *Chromista*, and group *Oomycota* (Dick 2002). Even though this is the most appropriate classification, this pathogen is still often referred to as a fungus (Rossman and Palm 2006).

The pathogen life cycle

P. viticola is an obligate biotrophic pathogen with *Vitis* spp. as hosts, among which *Vitis vinifera* is highly susceptible to infection. This heterothallic microorganism has dimorphic reproductive forms, with both sexual and asexual spores. Oospores, which represent the sexual stage of *P. viticola*, are formed in host leaves by the fusion of an oogonium and antheridium (Wong et al. 2001; Berlese 1898). Oospores usually form within the infected leaf tissue (Arens 1929) in the so called "mosaic leaves" (Fig. 2a) from the time of berry ripening to leaf fall. Oospores overwinter in the leaf litter on the soil or buried in the soil (Galbiati and Longhin 1984; Gehmann 1987). Several studies have focused on oospore development during the overwintering period (Burruano et al. 1992, 1987; Vercesi et al. 1999a; Rossi and Caffi 2007). During winter, the oospores reach morphological maturity, i.e., the inner oospore wall becomes thick, the nuclei merge, an ooplast is formed, and, finally, large lipid globules break into smaller ones (Vercesi et al. 1999a). Even if oospores are morphologically mature and even if environmental conditions are favourable for germination, oospores germination may be prevented by dormancy (Galet 1977). Spore dormancy is a general phenomenon that is regulated by the environment, nutrient permeability and endogenous inhibitors (Schweizer and Oliver 1999). When oospores dormancy is broken, they are said to be physiologically mature, i.e., they are able to start the germination process if environmental conditions are favourable (Ronzon and Clerjeau 1989). Rossi and Caffi (2007) recently demonstrated that leaf litter moisture and temperature regulate oospore development. The physiological maturation of the overwintering oospores was accurately described by the "hydrothermal time", which expresses the rate of physiological development based on a temperature- and moisture-dependent function (Rossi et al. 2008a). Other recent studies showed that calcium from both extracellular compartments and intracellular reserves may also affect the development of oospores (Vercesi et al. 2010).

When dormancy is broken, oospores are able to germinate. However, because the oospores are formed over a long period of time during the downy mildew epidemic, the oospore populations are not coeval, and

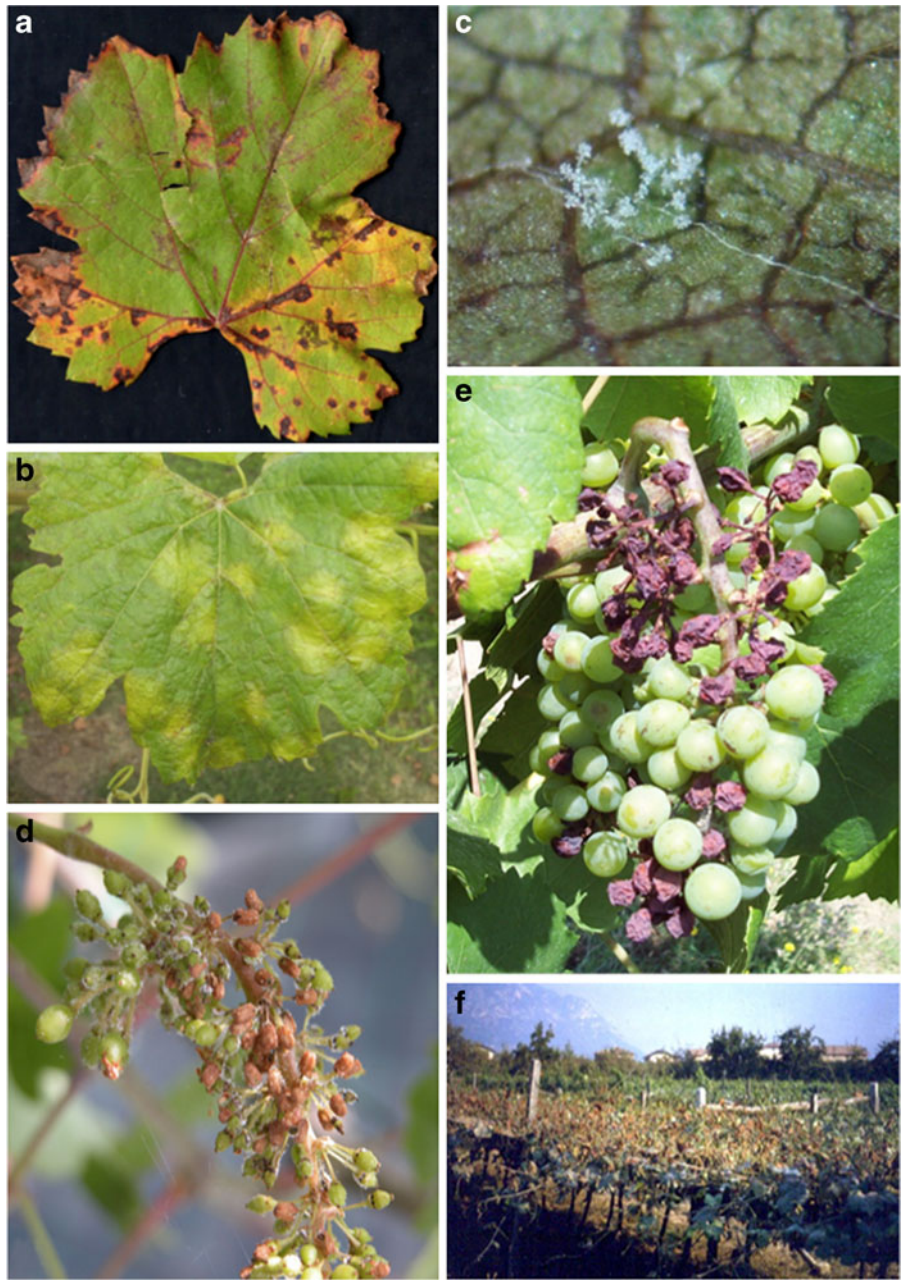


Fig. 2 Disease symptoms caused by *Plasmopara viticola* on *Vitis vinifera* L.: mosaic symptoms on an old leaf (oospores typically form in older leaves) (**a**); pale yellow, undefined oil spots represent the classic symptom that characterizes the disease on leaves (**b**); tree-shaped sporangiophores (magnification 10×) emerge from stomata on the abaxial side of the leaf, beneath the oil spots, (**c**); on young bunches with functional stomata, *P. viticola* infects both the rachis and berries and produces abundant sporulation, and the bunch appears "S" shaped (**d**); after fruit set, sporangiophores can no longer emerge from stomata, and the pathogen causes the dehydration of berries (**e**); downy mildew epidemics can cause severe yield losses because of direct effects on both bunches and leaves as well as the indirect effect via defoliation (**f**)

hence germination occurs gradually during the current season (Jermini et al. 2003; Rossi et al. 2008a) and even in the following seasons (Arens 1929). A recent study demonstrated that *P. viticola* oospores remained viable for at least 65 months after they were produced (Caffi et al. 2011a). Germination is faster when the temperature is between 20 and 24 °C and when there is rainfall or when the leaf litter (or the soil) that contains the oospores is moist because of rainfall or water flux from the atmosphere (Rossi et al. 2008a). A Gompertz equation calculated over the hydrothermal time correctly described the seasonal germination dynamics of *P. viticola* oospores (Rossi et al. 2008a).

Germinated oospores form sporangia that survive from a few hours to few days, depending on temperature and humidity (Blaeser and Weltzien 1979). Viable sporangia release four to eight zoospores in water (Muller and Sleumer 1934; Unger et al. 2007). *P. viticola*

zoospores are biflagellate, and they must keep swimming in water to survive (Arens 1929); they are sensitive to dryness and rapidly die when water dries (Muller and Sleumer 1934; Kiefer et al. 2002). Viable zoospores on the soil are dispersed to grape leaves by rain splash (Galbiati and Longhin 1984; Rossi and Caffi 2012). Recent studies demonstrated that even a relatively light rainfall (<0.2 mm/h) can disperse zoospores from soil to leaves (Rossi and Caffi 2012). Once deposited on the leaves, zoospores swim in the water film on the abaxial leaf surface towards stomata; when near stomata, they shed their flagella (Kiefer et al. 2002), form a germ tube that penetrates the stomata, and encyst in the substomatal cavity (Unger et al. 2007). The encysted zoospores produce hyphae that colonize the host tissue and the initial disease symptoms are noticed at the end of an incubation period as "oil spots" in the infected leaves (Fig. 2b).

The length of the incubation period is influenced by temperature and relative humidity (Goidanich et al. 1957). This relationship has been studied in the past and was recently re-visited by Rossi et al. (2002) in order to calculate the hourly progress of incubation and its confidence interval (Rossi et al. 2008b). Length of the incubation period also depends on both the target organ of the host and on ontogenic, or age-related, resistance (Rho et al. 2004; Develey-Rivière and Galiana 2007); the length of the *P. viticola* incubation period in berries in longer than in leaves (Rho et al. 2004).

Five to 18 days after infection has occurred, *P. viticola* produces sporangiophores bearing sporangia (Fig. 2c). Sporangia contain asexually produced zoospores (Lalancette et al. 1988a) that are the inoculum responsible for secondary infections (Lalancette et al. 1988b). Sunlight inhibits *P. viticola* sporulation (Brook 1979), hence sporulation occurs at night (Muller and Sleumer 1934; Yarwood 1937; Rumbolz et al. 2002). Night also usually provides high relative humidity (RH), which is another requirement for sporulation. Under controlled conditions, a minimum of 93 to 98 % RH was needed for sporangia production (Blaeser and Weltzien 1978; Leu and Wu 1982), but sporulation was observed in vineyards when RH ranged from 65 to 100 % (Istvantti and Palinkas 1913; Arens 1929; Muller and Sleumer 1934). Temperature is not considered a limiting factor for sporulation under field conditions (Gregory 1915; Richards 1959): *P. viticola* sporulation was observed on American varieties at temperatures ranging from 10 to 30 °C, with an optimum at 20 °C (Lalancette et al. 1988b). On *V. vinifera*, sporulation was highest when the temperature ranged from 18 to 20 °C (Gregory 1915) and from 16 to 28 °C (Leu and Wu 1982). Caffi et al. (2012) observed sporulating lesions at temperatures between 10.6 and 32.9 °C. Blaeser (1978) and Zachos (1959) found 13 °C to be the minimum constant temperature for sporulation. Sporulation was also observed in vineyards when temperature was 10 to 12 °C (Arens 1929; Muller and Sleumer 1934). The sporulation process has never been observed to begin with a temperature lower than 12.5 °C, but once initiated, sporulation can be completed even at a temperature as low as 9.5 °C (Hill 1989).

Downy mildew epidemics cause both direct and indirect yield losses. These losses result from rotting of inflorescences (Fig. 2d), shoots, and clusters (Fig. 2e), from the reduction in photosynthetic activity of the affected leaves, and from the premature defoliation of vines (Fig. 2f) (Caffi et al. 2006).

Kennelly et al. (2005) showed that stomata on berries of some *V. vinifera* varieties quickly become physiological inactive and behave as lenticels, thereby preventing penetration into the berry and sporulation from the berry. Even though pedicels remained susceptible to infection, the berries of some *V. vinifera* varieties exhibit a high level of ontogenic resistance starting about 3 weeks after bloom (Kennelly et al. 2005). Generally, the susceptibility of leaves to infection declines as leaves age (Gehmann 1987).

Disease epidemiology

Until recently, *P. viticola* oospores were considered as only important for initiating the disease early in the grapevine vegetative season but not for subsequent disease development. The explosive development and spread of the disease was attributed to the sporangia, which were produced by asexual reproduction and which were assumed to migrate long distances within short times. Several studies conducted in the 2000s had the objective to investigated these assumptions (Gobbin et al. 2003a) and, specifically, the qualitative and quantitative contribution of oosporic (sexual) versus clonal (asexual) inocula (Gessler et al. 2003; Gobbin et al. 2005) as well as the mode of disease spread (Gobbin et al. 2003b; Rumbou and Gessler 2006). As described below, a combination of research on epidemiology and

population genetics has substantially altered our understanding of downy mildew epidemics (Rumbou and Gessler 2004; Gobbin et al. 2006).

Gobbin et al. (2006) surveyed naturally-occurring downy mildew epidemics in 39 vineyards in 2000, 2001, 2002, and 2004 in Europe. The vineyards were located in Germany (n=4 vineyards), France (n=4), Italy (n=12), Greece (n=10), and Switzerland (n=9). Within each vineyard and growing season, the survey period started when the first lesions were noticed and ended when downy mildew generated a mosaic pattern, which prevented the collection of distinct lesions. On average, the survey period lasted 1.5 months and ended in July or August. Leaf pieces with oil spots was collected on two to 22 sampling dates per vineyard per year throughout the 39 vineyards; over all vineyards and sampling dates, 155 samplings were performed. About 10,000 oil spots were collected, and the oomycete strain associated with each oil spot was genotyped with the use of four specific microsatellite markers. Because *P. viticola* is diploid, every oil spot was characterized by a genetic profile consisting of eight microsatellite alleles. Genotypes presenting the same allele pattern were considered as clones (derived from the same oospore through asexual reproduction), while those presenting a different allele pattern were considered to have been derived from different oospores. Population genetics analyses were conducted to describe population structures (patterns of genotypic diversity, genetic variation, migration, genetic drift, etc.) and to assess genetic distances among populations. In parallel, a spatial distribution analysis of the epidemics was conducted (Gobbin et al. 2006).

In many of the 39 vineyards in central Europe, Gobbin et al. (2006) found that the first lesions in the epidemic (i.e., the primary infections caused by oospores) were often very abundant and that the pathogens in the lesions were genetically diverse. The abundance of primary infections and the high genotypic diversity detected early in the season (before the first lesions sporulated) indicated that the overwintering oospores in the soil constituted a very large and very diverse inoculum pool. In addition, lesions with new genotypes were detected throughout the growing season, indicating that oospores continued to germinate and cause primary infections throughout the growing season. These findings showed that at least for some vineyards, the assumptions that oospores cause only a small number of infections and only at the start of the epidemic were wrong. As discussed later in this section, however, epidemics in some vineyards were consistent with the traditional assumptions.

Gobbin et al. (2005 and 2006) indicated that the role of oospores and primary infections had been underestimated in many vineyards and the contribution of secondary (asexual) cycles to downy mildew epidemics had generally been overestimated. Most genotypes in each population (85 %, on average) were identified only once or twice throughout the survey period; this high genotypic diversity is inconsistent with a high level of clonal reproduction. In most vineyards, only one or two genotypes (<1 % of all genotypes) per epidemic underwent significant asexual reproduction. The contribution of dominant genotypes to the total disease severity ranged from 4 % to 95 %, depending on the epidemic surveyed, the macro-and microclimatic conditions, and the epidemic stage. The quantity of clonal infections was low until the middle of July and increased only in late summer. Analysis of the spatial distribution of genotypes within a vineyard showed that secondary infections were spatially aggregated in most cases (at the leaf, branch, or vine level). In one asexual cycle, clone dispersal was usually limited to a few vines around the site where the oosporic infection was first identified. Migration distances were generally less than 20 m per asexual cycle (Fig. 3) but were estimated to be up to 130 m in one case in Freiburg, Germany (Gobbin et al. 2005). With an increasing number of asexual cycles later in the growing season, the clones' spread following a stepwise migration pattern. Although there were exceptions (as discussed later in this section), the overall finding was that clonal infections in central Europe and the Greek mainland can be substantial in small parts of vineyards (delimited, for instance, by a few vines) and moderate at the level of the entire vineyard. Asexual inoculum, however, is much less important for downy mildew epidemics than previously believed.

Thus far, this section has emphasized a pattern of epidemic development in which primary infections are often more important than secondary infections. This pattern was found in the majority of the central European vineyards and in the mainland vineyards of Greece. However, a second pattern was also found, and this pattern was consistent with the traditional assumption regarding the relative importance of primary and secondary infections. In this second pattern,

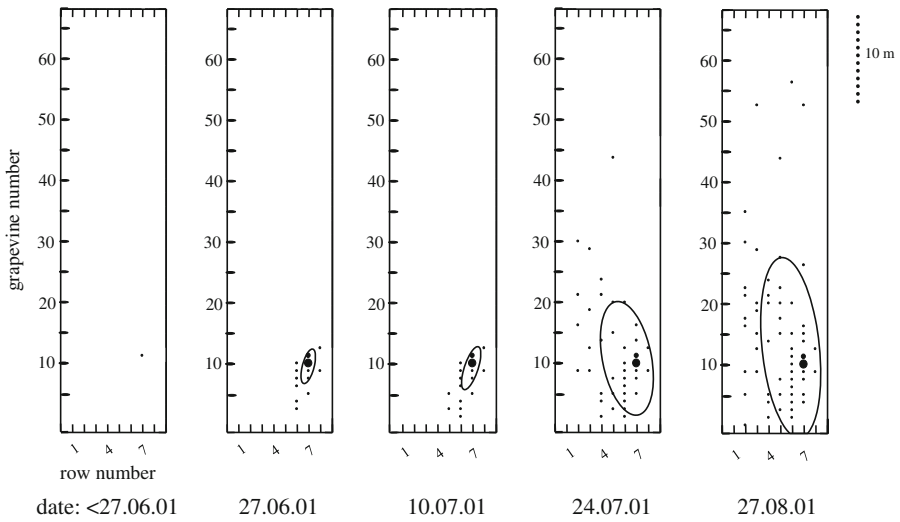


Fig. 3 Cumulative spatio-temporal dispersion pattern of the dominant genotype of *Plasmopora viticola* on *Vitis vinifera* in Bommes, France, in 2001. The plot is represented by a scaled rectangle (distances are shown by the dotted line on the right side). Vertical x-axis ticks describe the beginning and the end of a grapevine row. Numbers on the x-and y-axis represent grapevine coordinates in the plot (row and grapevine number, respectively). Dot size is proportional to the natural logarithm of the cumulative number of dominant genotype lesions per vine. Gaussian bivariate confidence ellipses, centred on the sample means of x and y lesion locations, were specified by a default probability value of 0.68. From left to right, plots describe the spread of the dominant genotype as a function of time (reprinted from Gobbin et al. 2005)

epidemics were characterized by a small number of primary infections and the predominance of only one or a few clones; this pattern occurred in low-severity epidemics in central Europe as well as on Greek islands and in coastal vineyards. In these cases, the small number of primary infections and the low genetic diversity of the pathogen indicated that the epidemics largely resulted from secondary cycles, in agreement with traditional understanding of downy mildew. On some islands (Rumbou and Gessler 2006) and in some isolated mountain regions (Gobbin et al. 2003b), the lesions derived from the dominant genotypes represented nearly 95 % of the sample size. The most dramatic increase in disease was observed in Freiburg (D): disease incidence resulting from secondary infections increased abruptly from 0 to 99 % on vines and from 0 to 45 % on leaves during 1 week in June 2004. Vines bore between 45 and 60 lesions each, corresponding to about 220,000 lesions/ha in a non-aggregated distribution.

The presence or absence of a bottleneck (i.e., a period of unfavourable weather conditions) was another feature of the epidemics studied. In cases where a bottleneck did not occur, the disease started in spring, increased continuously due to both primary and secondary infections, and ended in late autumn. This pattern was found in central Europe and in one Greek vineyard. In the remainder of the Greek vineyards, in contrast, the disease started in spring, increased until mid-summer, and then stopped because both primary and secondary infections were prevented by unfavourable weather conditions (heat and drought). The disease re-started in autumn when conditions became favourable again, resulting in a two-peak epidemic; the infections in autumn were evidently caused by oospores because no asexual inoculum survived the heat and drought.

The different contributions of primary versus secondary infections in combination with the presence or absence of bottlenecks during an epidemic led to different genetic substructures among samples within the same vineyard. During a single grape-growing season, either one (typical for "continuous" epidemics in central Europe) or two (typical for "two-peak" epidemics in Greece) *P. viticola* subpopulations were responsible for the epidemic. Among samples of two or more consecutive grape-growing seasons, either one or more subpopulations were responsible for the epidemics. The first case occurred only in two Greek vineyards characterized by the predominance of one clone and low disease severity, while the second case was more common and was found in all plots where the epidemic pattern was of the "two-peak" type.

Genetic subdivision among oosporic populations from different vineyards was very clear (Fig. 4). Most pair-wise comparisons between populations from geographical sites separated by more than 5 km revealed a significant genetic differentiation. This means, in biological terms, that the naturally occurring exchange of propagules (oospores or sporangia) between vineyards was low. Furthermore, the widely held belief that secondary sporangia disperse long distances, causing the homogenisation of the *P. viticola* population in Europe, was not supported by the data. Significant isolation by distance was found in central European *P. viticola* populations, suggesting a stepwise migration model (Gobbin et al. 2005, 2006). Conversely, in Greece no significant correlation was found between geographical distance and genetic divergence: pairs of population in Greece that were separated by an increasing geographical distance exhibited an increasing genetic divergence in some cases but not in others. The latter phenomenon is most probably due to the frequent natural geographic barriers such as the sea and the mountains, as well as the frequent population bottlenecks occurring in these populations. Such geographic barriers and population bottlenecks limit natural migration among populations.

All these studies show that oospores play a key role in the development of downy mildew epidemics: in addition to triggering them, they actively contribute to the progress of epidemics during a part of the growing season when both sexual and asexual spore types occur. To better understand the epidemics caused by pathogens with dimorphic spores with repeated primary and secondary infection cycles, a simulation model was recently elaborated and applied to the grapevine downy mildew system (Rossi et al. 2009). This theoretical model represents the conceptual framework in which mechanistic models simulating the primary and secondary infection processes of *P. viticola* can be linked together to obtain a dynamic simulation of the entire downy mildew epidemic on grapevine. The

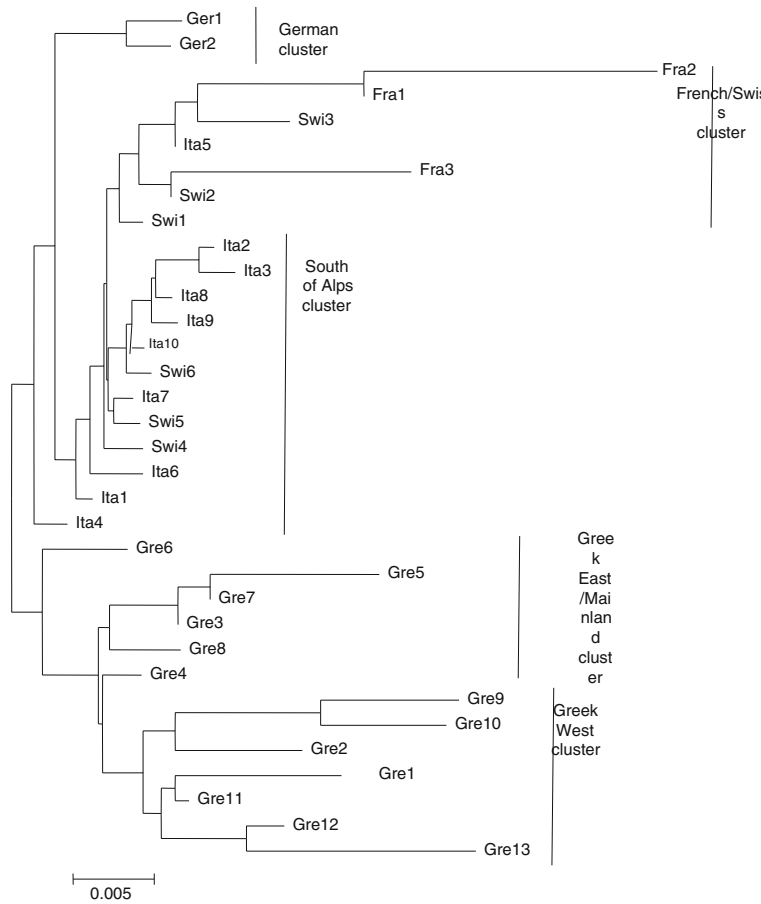


Fig. 4 Unrooted neighbour-joining tree based on Shriver's genetic distance D_{sw} among 34 European *Plasmopara viticola* populations. Origins of populations are: France (Fra), Germany (Ger), Greece (Gre), Italy (Ita), and Switzerland (Swi) (Reprinted from Gobbin et al. 2006)

model accurately reproduced the variability of the real system. The model mimicked both those downy mildew epidemics characterized by a high degree of clonality (with most seasonal lesions caused by a few dominant genotypes) and those with a prevalence of oosporic-derived lesions (Fig. 5).

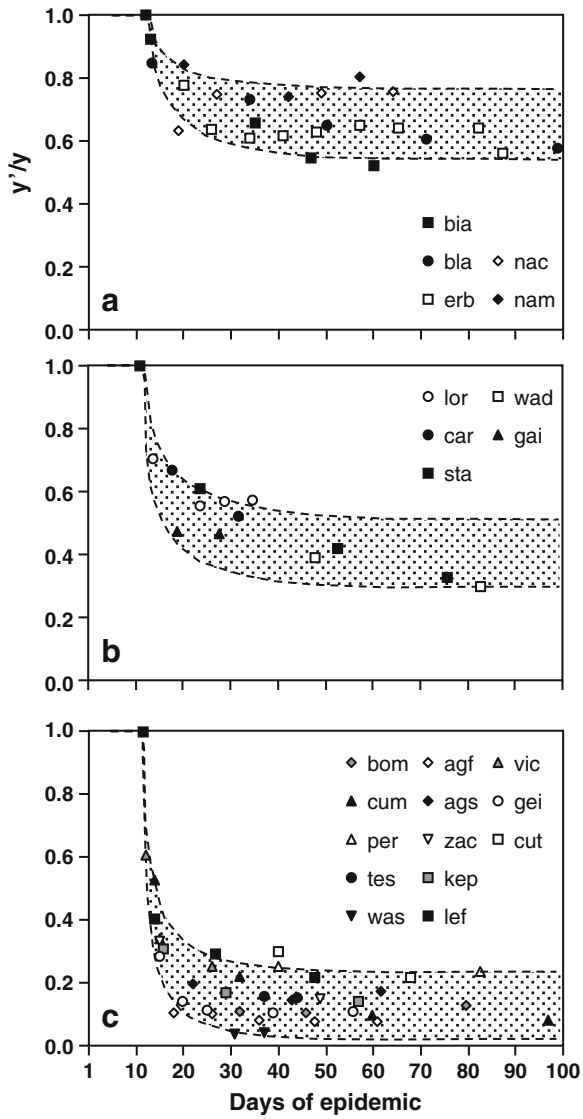


Fig. 5 Comparison between the relative contribution of primary downy mildew lesions to total lesions (ratio y'/y) simulated by the model and observed in 23 vineyards with epidemics characterized by low (**a**), intermediate (**b**), and high (**c**) level of clonality. The lower and upper curves in each panel indicate the 99 % confidence intervals of predicted outcome (shaded area). Symbols and acronyms are different vineyards (reprinted from Rossi et al. 2009)

Relationships between weather and downy mildew

The existence of relationships between environmental factors, host susceptibility, and *P. viticola* has long been recognized: the so-called 3–10 rule was the first attempt to predict primary infections caused by *P. viticola* (Baldacci 1947). This rule states that primary infections are likely when the following conditions occur simultaneously: (1) air temperature is ≥10 °C; (2) vine shoots are at least 10 cm long; and (3) at least 10 mm of rain has fallen during the last 24–48 h.

Other weather driven models have been developed in France (Stryzik 1983; Tran Manh Sung et al. 1990), Germany (Hill 1990, 2000), the USA (Park et al. 1997), Italy (Orlandini et al. 1993; Rossi et al. 2008b), and Australia (Magarey et al. 1991) by either empirical or mechanistic approaches. Empirical models are elaborated starting from data collected under specific field conditions, and they do not necessarily account for cause-and-effect relationships between variables; mechanistic models are based on a priori analysis of the factors influencing epidemics (Caffi and Rossi 2009). Weather driven, mechanistic models explain the observed behaviour of the system on the basis of what is known about how the system works in relation to the weather variables, which represent the main model inputs (Rossi et al. 2010). A detailed study was recently carried out to compare the different modelling approaches that have been used for simulating grapevine downy mildew primary infection (Caffi et al. 2007). The empirical models were found to be too over-simplified to correctly simulate such a complex biological phenomenon. In fact, empirical models fail to consider some key elements of the infection process. Empirical models also simultaneously simulate various biological events that are not simultaneous and that are differentially influenced by specific weather conditions. This is the case for the 3–10 rule (Baldacci 1947), which was derived from field observations and which fails to consider oospore maturation or the different stages of the infection process.

Another problem with the empirical approach is that it is difficult to use an empirical model under environmental conditions that differ from the conditions occurring during model development. The EPI model (Stryzik 1983), for instance, is based on the assumption that the pathogen has been differently adapted to climatic conditions of different grape-growing area: this empirical model was elaborated in

the Bordeaux area (France), where it has reliably predicted downy mildew epidemics. When used under different climatic conditions, however, the EPI model required specific calibrations and modifications that make its use in disease prediction difficult (Vercesi et al. 1999b; Vercesi and Liberati 2001; Caffi et al. 2007). Another model, DMCast (Park et al. 1997), which was elaborated in New York State (USA), overcomes the above problems because of its mechanistic approach but uses an empirical distribution of oospore maturation; this empirical distribution requires validation and calibration before the model can be used in different environments (Caffi et al. 2007). A recent model developed at Piacenza (Italy) with a fully mechanistic approach (Rossi et al. 2008b) lacks these limitations; it does not require calibration or correction and provides an accurate, detailed, and dynamic simulation of the sexual stage of *P. viticola* (Caffi et al. 2007; Caffi et al. 2009).

Mechanistic modelling of primary infections

The above-mentioned mechanistic model for primary infections of *P. viticola* was elaborated using "system analysis" (Rossi et al. 2008b). The life cycle of *P. viticola* was divided into different stages, which represent the state variables of the model. The model assumes that oospores in the affected leaves of the previous seasons form the seasonal oospore dose and that these oospores are all morphologically mature by the end of the previous season. The model simulates the physiological maturation of the oospores according to the hydrothermal time (Rossi et al. 2008a). When the oospores become physiologically mature (i.e., when the oospores break dormancy), the model assumes that each rainfall (≥0.2 mm/h) triggers the germination process of oospores in the leaf litter or soil. Once the germination process is triggered, oospore germination is simulated according to temperature (Laviola et al. 1986) when humidity is not a limiting factor (Rossi and Caffi 2007). Germinated oospores form sporangia that can survive for a certain period of time depending on temperature and relative humidity (Blaeser and Weltzien 1979). After simulating oospore germination, the model simulates zoospore ejection based on the presence of leaf wetness and temperature (Ravaz 1914; Galet 1977). At this stage, zoospores are swimming in water films on the leaf litter or on the soil and survive there as long as the water films persist. The model assumes that when viable zoospores are present, any rainfall can splash zoospores onto grape leaves where, if susceptible leaf tissue is present, they cause infection depending on a combination of favourable conditions of temperature and wetness duration (Blaeser and Weltzien 1979). Finally, oil spots appear on leaves at the end of a temperature-dependent incubation period (Goidanich et al. 1957). Therefore, this model takes into account all the stages of the primary infection cycle: oospore maturation and germination; survival of zoosporangia; zoospore ejection, survival, and dispersal; and infection and incubation. The dynamic simulation allows researchers and pest managers to follow the infection process step-by-step and to estimate the period of disease symptom onset.

The model was evaluated in more than 100 vineyards in Northern, Southern, and Insular Italy (1995 to 2007) as well as with potted grapevine plants exposed to inoculum (2006 to 2008) (Caffi et al. 2009). The model always showed high sensitivity (>0.99) and specificity (0.87). Overall accuracy was 0.90, and the likelihood ratios of predicted infections and non-infections also showed good accuracy (Madden 2006). Nevertheless, a false positive proportion existed, i.e., the model predicted an infection but the disease did not appear in 13 % of the cases (Caffi et al. 2009).

This model was also evaluated under the environmental conditions in the Province of Quebec, Eastern Canada, by comparing the time of lesion emergence predicted by the model with field observations (Caffi et al. 2011b). Without any calibration or modification of the algorithms, the model performed as well under Canadian conditions as it had under Italian conditions. In particular, all observed lesion emergences except for one were correctly predicted by the model, giving a true positive proportion of 0.996, a true negative proportion of 0.91, a false positive proportion of 0.004, and a false negative proportion of 0.09. Overall model accuracy was 0.95 (Caffi et al. 2011b).

Thus, the model accuracy was high in very different environments. Although the model generated some false positive predictions, most of these occurred when grapevines were between bud break and the 5-to-6 leaves unfolded stage (about 10 cm shoot length), and thus the host surface area available for infection was very low. Some incorrect model predictions that occurred in early-season could be avoided by accounting for the

grape growth stage at the time when infection is predicted, or by linking the downy mildew model to a grape phenological model (Ortega-Farías et al. 2002; Mariani et al. 2007).

Mechanistic modelling of secondary infections

Models were also developed to simulate the progress of downy mildew epidemics due to infections caused by asexual sporangia of *P. viticola*, starting from when the first disease lesions are observed in a vineyard or forecast by the 3–10 rule or similar rules. These models include VINEMILD (Blaise and Gessler 1992), MILVIT (Magnien et al. 1991), D-model (Magarey et al. 1991), PRO (Hill 1990), and PLASMO (Orlandini et al. 1993), which were developed in Switzerland, France, Austria, Germany, and Italy, respectively. All these are compartmental models that work in a relatively similar way and provide a prediction of disease severity or a risk index that guides the scheduling of fungicide applications.

For instance, PLASMO simulates the main processes of *P. viticola* secondary infections in order to calculate the disease severity on leaves (Orlandini et al. 1993). The infection stages considered are: oil spot survival, sporulation, spore survival, inoculation, and incubation. Using weather data, PLASMO estimates the occurrence of these stages and the progress of the incubation period whenever an infection is predicted. The model output is the percentage of leaf area covered by downy mildew lesions. The model was empirically calibrated with data collected over 2 years in vineyards of Tuscany (Central Italy); the empirical data were used to tune the coefficient of the multiplication of the number of spores per leaf area and the coefficient of spore survival. PLASMO was validated against independent data collected from 1998 to 2003 in the same vine-growing area, and correctly predicted 66.6 % of the data (Orlandini et al. 2008). One possible cause for the incorrect predictions is that the model was initialized by using the severity of the first disease symptoms observed in the vineyard, which is difficult to detect correctly because the first seasonal disease symptoms usually appear with a very low incidence (Hill and Breth 1997) and are randomly distributed in the vineyard (Madden et al. 1995). This initial assessment might be improved if the observers stratified the vineyards into areas of different disease risk and searched those parts of the vines with highest risk of disease occurrence (Seem et al. 1985); the part of the vine with the highest risk of disease is the lower canopy layer, where the probability of impact from rain-generated splashes is greatest (Rossi and Caffi 2012).

Moreover, a recent study demonstrated that i) at least 3 wet hours in the night is a useful predictor for frequency of sporulation of *P. viticola*, ii) once sporulation started sporangia are produced at a high rate for the first 4 days and then slow down until sporulation stops, iii) the total production of sporangia increases with humidity of the sporulation period and, finally, that iv) sporangia are released over a wide range of weather conditions, including dry periods (Caffi et al. 2012). All this new, relevant information should be helpful in refining epidemiological models dealing with the secondary infection of grapevine downy mildew.

Disease control

In 1882, Alexis Millardet discovered the protectant effect of calcium hydroxide and copper against the devastating epidemics of *P. viticola* and developed 'Bordeaux mixture', which became the first fungicide used in viticulture (Delmotte et al. 2006). Developments in downy mildew control have been recently reviewed by Gessler et al. (2011). Currently, downy mildew can be controlled with properly timed fungicide applications, and disease control programs usually focus on controlling primary infections in the pre-bloom and early post-bloom of vines and on limiting secondary spread during the summer. In Northern Italy, for instance, six to nine fungicide sprays are used to control the disease (Borgo et al. 2004), with an estimated cost of 30 € ha^{-1} $spray^{-1}$ (Monchiero et al. 2005). Across the entire Piedmont region, the annual cost for downy mildew control typically ranges from 8 to 16 million €, depending on weather conditions (Salinari et al. 2006). However, within-season variations in weather conditions often result in both unexpected disease outbreaks and the failure of predicted epidemics to occur, indicating the need for an improved prediction and warning system.

Toward a disease warning system

Some of the previously mentioned disease models have been used to improve fungicide scheduling in the area where the model was developed, but none of

them is sufficiently accurate and robust for use in other areas, including Italy (Egger et al. 1994; Vercesi et al. 1999a; Vercesi and Liberati 2001). As a consequence, disease warnings throughout Italy continue to be based on the simple 3–10 empirical rule (Rossi et al. 2000), although this rule often predicts infection periods that do not occur (Serra et al. 1998; Vercesi et al. 1999a). Recently, the model of Rossi et al. (2008b) was evaluated in a real-time warning system for downy mildew control (Caffi et al. 2010). The warning system includes three components: i) weather data and 3-day forecasts; ii) the disease model; and iii) a mobile phone short-message service (SMS). When the model determined that there had been an infection period (based on weather data) or that there would be an infection period (based on predicted weather forecasts), an SMS was sent to the mobile phone of the managers of six experimental vineyards. Two kinds of SMSs were sent: one stated that "a downy mildew infection period is predicted for a future day" and the other stated that "a not previously predicted infection period occurred on the previous day".

In the six experimental vineyards, an unsprayed control was compared with treatments applied whenever the warning system predicted an infection period and with treatments applied according to the grower's standard schedule. High levels of disease control were obtained with both kinds fungicide treatments (based on the model and on the grower's schedule) when disease severity was either high or moderate; when disease severity was very low, the disease control was always complete. Average efficacy in reducing disease severity was greater than 90 % for both treatments. Even though the number of sprays varied considerably among experiments (i.e., 1 to 5), use of the warning system reduced the number of sprays from 33 to 86 % in the six experiments, with a median reduction of 54 %. The grower's spray schedule was the most expensive control strategy, averaging 337 €/ha, with a minimum of 262 €/ha in 2006 and a maximum of 467 €/ha in 2008. The treatments based on the disease warning system were less costly, averaging 164 €/ha (range of 52 to 260 €/ha). The average saving was 174 €/ha, with the highest saving of 311 €/ha in 2007, when the disease did not develop. Similar results have been confirmed in other studies (Bortolotti et al. 2012; Morroni et al. 2012).

The use of the warning system, then, substantially reduced fungicide applications and associated costs. Environmental benefits were also recorded: each application that was avoided represented a reduction of 4.9 l/ha in fuel consumption and a related reduction of about 12.5 kg/ha in CO_2 emissions (Caffi et al. 2010). The system is currently included in the warning services provided to growers by Plant Protection Organisations in several areas of Italy and in two Decision Support Systems, one for organic growers (named Vitebio.net™) (Caffi et al. 2011c) and one for growers following IPM (Integrated Pest Management) guidelines (named Vite.net™) (Salinari et al. 2012).

Future perspectives

Molecular studies carried out over the last decade have greatly increased our understanding of the epidemiology of grapevine downy mildew. More than a five-year period was necessary to assemble and integrate the independent results from different research groups working on molecular differences between primary and secondary lesions in North America, Europe, South Africa, and Australia. The view that a downy mildew epidemic is almost exclusively based on the asexual reproduction of the pathogen, with the sporangia able to migrate for long distances in massive amounts, has been progressively abandoned. The view increasingly accepted is that an epidemic of *P. viticola* is caused by a very large number of different genotypes as represented in the pool of oospores; few of these genotypes are able to establish progeny via asexual sporangia. The new view clarifies the dynamics of many epidemics that were previously inexplicable. One such case was the onset of the disease in late season in a vineyard where no primary infection had been observed. This was difficult to explain because, according to the traditional view, substantial disease in late season should be caused by secondary infections, which cannot occur without primary infections. The explanation for this example is that some vineyards contain a very large pool of oospores, which may not germinate until conditions are suitable and which can cause a substantial level of disease. The traditional view of *P. viticola* epidemiology is only valid in those cases when there is a dominant genotype, i.e., one that is able to cause more than 50 % of the total infections. The recent findings have also shown how the diversity of this pathogen is the key to its adaptability to different micro-climates, different host varieties, and different cropping systems worldwide.

These new findings have also influenced the development of new disease prediction models (Rossi et al. 2008b) and consequently the development of new warning systems (Caffi et al. 2010) for improved management of downy mildew in the vineyard.

Other new information concerning the biology of *P. viticola* is emerging, as researchers are currently studying the dynamics of oospore formation within infected tissues, oospore survival across seasons, and oospore numbers in the litter and soil. This new information, when combined with the other recent findings concerning oospore maturation, germination, and contribution to epidemics, should help in the development of new strategies for sustainable management of downy mildew in vineyards.

References

Arens, K. (1929). Untersuchungen über Keimung und Zytologie der Oosporen von *Plasmopara viticola* (Berl. et De Toni). *Jahrbucher Wissenschaftliche Botanik, 70*, 57–92.

Baldacci, E. (1947). Epifitie di *Plasmopara viticola* (1941–46) nell'Oltrepó Pavese ed adozione del calendario di incubazione come strumento di lotta. *Atti Istituto Botanico, Laboratorio Crittogamico, 8*, 45–85.

Berlese, A. N. (1898). Cytological studies on the fungi. The division of the nucleus and formation of conidia in Oidium. *Rivista di Patologia Vegetale, 6*, 66–75.

Blaise, P., & Gessler, C. (1992). Vinemild: toward a management tool for grape downy mildew. *Acta Horticulturae, 313*, 257–262.

Blaeser, M. (1978). Untersuchungen zur Epidemiologie des falschen Mehltaus an Weinrebe, *Plasmopara viticola* (Berk. et de Toni). PhD Thesis, Universität Bonn Germany, 127 pp. Tipo di documento non previsto nelle istruzioni della rivista

Blaeser, M., & Weltzien, H. C. (1978). The importance of sporulation, dispersal, and germination of sporangia of *Plasmopara viticola*. *Journal of Plant Disease and Protection, 3*, 155–161.

Blaeser, M., & Weltzien, H. C. (1979). Epidemiologische Studien an *Plasmopara viticola* zur Verbesserung der Spritzterminbestimmung. *Z Pflanzenk Pflanzen, 86*, 489–98.

Borgo, M., Bellotto, D., & Zanzotto, A. (2004). Composti a base di fenamidone contro la peronospora della vite. *L'informatore Agrario, 50*, 49–53.

Bortolotti, P. P., Nannini, R., & Bugiani, R. (2012). Applicazioni di campo del modello previsionale per peronospora della vite (*Plasmopara viticola*). *In Atti Giornate Fitopatologiche, 2*, 401–404.

Brooke, P. J. (1979). Effect of light on sporulation of *Plasmopara viticola*. *New Zealand Journal of Botany, 17*, 135–138.

Burruano, S., Conigliaro, G., & Ciofalo, G. (1992). Indagine preliminare sull'influenza dell'umidità del terreno sulla maturazione delle oospore di *Plasmopara viticola*. *Phytopathologia Mediterranea, 31*, 1–4.

Burruano, S., Strazzeri, S., & Laviola, C. (1987). Influenza dell'acqua sulla germinazione delle oospore di *Plasmopara viticola*. *Phytopathologia Mediterranea, 26*, 19–22.

Caffi, T., Rossi, V., Giosuè, S., & Poni, S. (2006). Effect of downy mildew infection on grape leaf photosynthesis. *Journal of Plant Pathology, 88*(3), 34.

Caffi, T., Rossi, V., Cossu, A., & Fronteddu, F. (2007). Empirical vs. mechanistic models for primary infections of *Plasmopara viticola*. *Bulletin OEPP/EPPO Bulletin, 37*, 261–271.

Caffi, T., & Rossi, V. (2009). L'evoluzione dei modelli epidemiologici per le infezioni primarie di *Plasmopara viticola*: dalla regola dei '3–10' ai modelli meccanicistici. In M. Severini & S. Pesolillo (Eds.), *Modelli per la difesa delle Piante* (pp. 59–80). Roma: Aracne Editrice srl.

Caffi, T., Rossi, V., Bugiani, R., Spanna, F., Flamini, L., Cossu, A., & Nigro, C. (2009). A model predicting primary infections of *Plasmopara viticola* in different grapevine-growing areas of Italy. *Journal of Plant Pathology, 91*(3), 535–548.

Caffi, T., Rossi, V., & Bugiani, R. (2010). Evaluation of a warning system for controlling primary infections of grapevine downy mildew. *Plant Disease, 94*, 709–716.

Caffi, T., Rossi, V., & Lusitani, M. (2011). Long-term survival of *Plasmopara viticola* oospores. *IOBC/WPRS Bulletin, 66*, 111–114.

Caffi, T., Rossi, V., & Carisse, O. (2011). Evaluation of a dynamic model for primary infections caused by *Plasmopara viticola* on grapevine in Quebec. *Plant Health Progress*. doi:10.1094/PHP-2011-0126-01-RS.

Caffi, T., Salinari, F., & Rossi, V. (2011). A decision support system for management of organic vineyards against downy mildew. *Phytopathology, 101*(6), S24.

Caffi,T., Gilardi, G., Monchiero, M. & Rossi, V. (2012). Production and dispersal of the asexual sporangia of *Plasmopara viticola*. *Phytopathology*, accepted for publication, http://dx.doi.org/10.1094/PHYTO-04-12-0082-R

Delmotte, F., Chen, W. J., Richard-Cervera, S., Greif, C., Papura, D., Giresse, X., Mondor-Genson, G., & Corio-Costet, M. F. (2006). Microsatellite DNA markers for *Plasmopara viticola*, the causal agent of downy mildew of grapes. *Molecular Ecology Notes, 6*, 379–381.

Develey-Rivière, M. P., & Galiana, E. (2007). Resistance to pathogens and host developmental stage: a multifaceted relationship within the plant kingdom. *New Phytologist, 175*, 405–416.

Dick, M. W. (2002). Towards and understanding of the evolution of the downy mildews. In P. T. N. Spencer-Phillips, U. Gisi, & A. Lebeda (Eds.), *Advances in downy mildew research* (pp. 1–57). Dordrecht: Springer.

Egger, E., Marinelli, E., & Greco, G. (1994). Un nuovo modello gestionale per il controllo della Peronospora della vite: prime verifiche nell'ambiente viticolo aretino. *Rivista di Viticoltura e di Enologia, 47*(1), 15–32.

Galbiati, C., & Longhin, G. (1984). Indagini sulla formazione e sulla germinazione delle oospore di *Plasmopara viticola*. *Rivista di Patologia Vegetale, 20*, 66–80.

Galet, P. (1977). Les Maladies et les Parasites de la Vigne. Tome I, *Le MaladiesDue à des Végétaux*. Montpellier: Le Paysan du. Midi.

Gehmann, K. (1987). Untersuchungen zur Epidemiologie und Bekämpfung des Falschen Mehltaus der Weinrebe, verursacht durch *Plasmopara viticola*. [German]. PhD thesis, University of Stuttgart-Hohenheim.

Gessler, C., Rumbou, A., Gobbin, D., Loskill, B., Pertot, I., Raynal, M., & Jermini, M. (2003). A change in our conception of the life cycle of *Plasmopara viticola*: Oosporic infections versus asexual reproduction in epidemics. *IOBC/WPRS Bulletin, 26*, 13–16.

Gessler, C., Pertot, I., & Perrazzolli, M. (2011). *Plasmopara viticola*: A review of knowledge on downy mildew of grapevine and effective disease management. *Phytopathologia Mediterranea, 50*, 3–44.

Gobbin, D., Jermini, M., Loskill, B., Pertot, I., Raynal, M., & Gessler, C. (2005). Importance of secondary inoculum of *Plasmopara viticola* to epidemics of grapevine downy mildew. *Plant Pathology, 54*, 522–34.

Gobbin, D., Pertot, I., & Gessler, C. (2003a). Identification of microsatellite markers for *Plasmopara viticola* and establishment of high throughput method for SSR analysis. *European Journal of Plant Pathology, 109*, 153–64.

Gobbin, D., Pertot, I., & Gessler, C. (2003b). Genetic structure of a *Plasmopara viticola* population in an isolated Italian mountain vineyard. *Journal of Phytopathology, 151*, 636–646.

Gobbin, D., Rumbou, A., Linde, C. C., & Gessler, C. (2006). Population genetic structure of Plasmopara viticola after 125 years of colonization in European vineyards. *Molecular Plant Pathology, 7*, 519–531.

Goidanich, G., Casarini, B., & Foschi, S. (1957). Lotta antiperonosporica e calendario d'incubazione. *Giornale di Agricoltura, 13*, 11–14. gennaio.

Gregory, C.T. (1915). Studies on *Plasmopara viticola*. (Official report of the session of *Internation Congress of Viticulture*, P.P.I.E. San Francisco).

Hill, G.K. (1989). Effect of temperature on sporulation efficiency of oilspots caused by *Plasmopoara viticola* (Berk. & Curt. ex De Bary) Berl. & De Toni in the vineyard. *Viticultural and Enological Sciences*, 44, 86–90.

Hill, G. K. (1990). Plasmopara Risk Oppenheim—a deterministic computer model for the viticultural extension service. *Notiziario sulle Malattie delle Piante, 111*, 231–223.

Hill, G. K., & Breth, K. (1997). Severe primary infections of *Plasmopara viticola*—A plant pathologists nightmare. *Viticulture and. Enological Science, 52*, 150–151. Non trovo la rivista per confermarne con sicurezza il titolo esteso!.

Hill, G. K. (2000). Simulation of *P. viticola* oospore-maturation with the model SIMPO. *IOBC/WPRS Bulletin, 23*(4), 7–8.

Istvantti, G., & Palinkas, G. (1913). Etudes sur le mildiou de la vigne. *Annales de l'Institut Central Ampelologique Royal Hongrois, 4*, 1–125.

Jermini, M., Gobbin, D., Blaise, P., & Gessler, C. (2003). Influence of the overwintering methods on the germination dynamic of downy mildew (*Plasmopara viticola*) oospores. *IOBC/WPRS Bulletin, 26*, 37–42.

Kennelly, M. M., Gadoury, D. M., Wilcox, W. F., Magarey, P., & Seem, R. C. (2005). Seasonal development of ontogenic resistance to downy mildew in grape berries and rachises. *Phytopathology, 95*(12), 1445–1452.

Kiefer, B., Riemann, M., Buche, C., Kassemeyer, H.-H., & Nick, P. (2002). The host guides morphogenesis and stomatal targeting in the grapevine pathogen *Plasmopara viticola*. *Planta, 215*, 387–393.

Lafon, R., & Clerjeau, M. (1988). Downy mildew. In R. C. Pearson & A. C. Goheen (Eds.), *Compendium of grape diseases* (pp. 11–13). St. Paul: APS Press.

Lalancette, N., Ellis, M. A., & Madden, L. V. (1988a). A quantitative model for describing the sporulation of *Plasmopara viticola* on grape leaves. *Phytopathology, 78*, 1316–1321.

Lalancette, N., Ellis, M. A., & Madden, L. V. (1988b). Development of an infection efficiency model of *Plasmopara viticola* on American grape based on temperature and duration of leaf wetness. *Phytopathology, 78*, 794–800.

Laviola, C., Burruano, S., & Strazzeri, S. (1986). Influenza della temperatura sulla germinazione delle oospore di Plasmopara viticola (Berk. et Curt.) Berl. et De Toni. *Phytopathologia Mediterranea, 25*, 80–84.

Leu, L. S., & Wu, H. G. (1982). Inoculation, sporulation and sporangial germination of grape downy mildew fungus, *Plasmopara viticola*. *Plant Protection Bulletin (Taiwan), 24*, 161–170.

Madden, L. V. (2006). Botanical epidemiology: some key advances and its continuing role in disease management. *European Journal of Plant Pathology, 11*, 3–23.

Madden, L. V., Hughes, G., & Ellis, M. A. (1995). Spatial heterogeneity of the incidence of grape downy mildew. *Phytopathology, 85*, 269–275.

Magarey, P. A., Wachtel, M. F., Weir, P. C., & Seem, R. C. (1991). A computer-based simulator for rationale management of grapevine downy mildew Plasmopara viticola. *Australian Plant Protection Quarterly, 6*, 29–33.

Magnien, C., Jacquin, D., Muckensturm, N., & Guillemard, P. (1991). MILVIT: a descriptive quantitative model for the asexual phase of grapevine downy mildew. *IOBC/WPRS Bulletin, 21*, 451–459.

Mariani, L., Failla, O., Dal Monte, G., & Facchinetti, D. (2007). *IPHEN: A model for real time production of grapevine phenological maps*. Zaragoza: Paper presented at Congress on Climate and Viticulture.

Monchiero, M., Gilardi, G., Garibaldi, A., & Gullino, M. L. (2005). Risultati di prove di lotta alla peronospora della vite in Piemonte. *Informatore Fitopatologico, 55*(4), 32–37.

Morroni, M., Zanzotto, A., Caffi, T., & Trevisol, F. (2012). Gestione dei trattamenti antiperonosporici mediante l'ausilio di un modello previsionale sulle infezioni primarie: due annate di sperimentazione nel trevigiano. *In ATTI Giornate Fitopatologiche, 2*, 393–399.

Muller, K., & Sleumer, H. (1934). Biologische Unterschungen über die Peronosorakrankheit des Weinstokes mit besonderer Breücksichtingung threr Bekämpfung nach der Inkhationskalendermethode. *Landwirtschaftliche Jahrbucher, 79*, 509–576.

Orlandini, S., Gozzini, B., Rosa, M., Egger, E., Storchi, P., Maracchi, G., & Miglietta, F. (1993). PLASMO: a simulation model for control of Plasmopara viticola on grapevine. *IOBC/WPRS Bulletin, 23*, 619–626.

Orlandini, S., Massetti, L., & Dalla Marta, A. (2008). An agrometeorological approach for the simulation of Plasmopara viticola. *Computers and Electronics in Agriculture, 64*(2), 149–161.

Ortega-Farías, S. O., Lozano, P., Moreno, Y., & León, L. (2002). Desarrollo de modelos predictivos de fenologìa y evoluciòn

de nadurez en vid para vino cv. *Cabernet Sauvignon y Chardonnay. Agricultura Técnica, 62*, 27–37.

Park, E. W., Seem, R. C., Gadoury, D. M., & Pearson, R. C. (1997). DMCAST: a prediction model for grape downy mildew development. *Viticultural and Enological Science, 52*, 182–189.

Ravaz, L. (1914), Traité général de viticulture. III partie: le mildiou. Broché, Montpellier, Paris 14, pp. 282–322.

Rho, G., Sancassani, G. P., Toffolatti, S., Fremiot, P., & Vercesi, A. (2004). Further investigations on the length of the incubation period of Plasmopara viticola (Berk. et Curt.) Berl. et De Toni. *Atti Giornate Fitopatologiche, 2*, 135–142.

Richards, C. E. (1959). A flexible growth function for empirical use. *Journal of Experimental Botany, 10*, 290–300.

Ronzon, T. M. S. C. & Clerjeau, M. (1989). Influence of climatic conditions in the autumn on the maturation of oospores of *Plasmopara viticola* oospores and the degree of attack of mildew in the spring: models of the phenomenon. Proceedings of a meeting of EC Experts' Group, Thessaloniki, Greece 6–8 October, 1987; 237–243.

Rossi, V., Ponti, I., & Cravedi, P. (2000). The status of warning services for plant pests in Italy. *IOBC/WPRS Bulletin, 30*, 19–29.

Rossi, V., Giosuè, S., Girometta, B., & Bugiani, R. (2002). Influenza delle condizioni meteorologiche sulle infezioni primarie di *Plasmopara viticola* in Emilia-Romagna. *Atti Giornate Fitopatologiche, 2*, 263–270.

Rossi, V., & Caffi, T. (2007). Effect of water on germination of Plasmopara viticola oospores. *Plant Pathology, 56*(6), 957–966.

Rossi, V., Caffi, T., Bugiani, R., Spanna, F., & Della Valle, D. (2008). Estimating the germination dynamics of Plasmopara viticola oospores using hydro-thermal time. *Plant Pathology, 57*(2), 216–226.

Rossi, V., Caffi, T., Giosuè, S., & Bugiani, R. (2008). A mechanistic model simulating primary infections of downy mildew in grapevine. *Ecological Modeling, 212*(3–4), 480–491.

Rossi, V., Giosuè, S., & Caffi, T. (2009). Modelling the dynamics of sexual and asexual spores during Plasmopara viticola epidemics. *Journal of Plant Pathology, 91*(3), 615–627.

Rossi, V., Giosuè, S., & Caffi, T. (2010). Modelling Plant diseases for decision making in crop protection. In E. C. Oerke, R. Gerhards, G. Menz, & R. A. Sikora (Eds.), *Precision crop protection—The challenge and use of heterogeneity* (p. 441). Dordrecht: Springer.

Rossi, V., & Caffi, T. (2012). The role of rain in dispersal of the primary inoculum of *Plasmopara viticola*. *Phytopathology, 102*(2), 158–165.

Rossman, A. Y., & Palm, M. E. (2006). Why aren't *Phytophthora* and other *Oomycota* true Fungi? *Outlooks in Pest Management, 17*, 217–219.

Rumbou, A., & Gessler, C. (2004). Genetic dissection of a *Plasmopara viticola* population from a Greek vineyard in two consecutive years. *European Journal of Plant Pathology, 110*, 379–392.

Rumbou, A., & Gessler, C. (2006). Particular structure of *Plasmopara viticola* populations evolved under Greek island conditions. *Phytopathology, 96*, 501–509.

Rumbolz, J., Wirtz, S., Kassemeyer, H. H., Guggenheim, R., Schafer, E., & Buche, C. (2002). Sporulation of *Plasmopara viticola*: differentiation and light regulation. *Plant Biology, 4*, 413–422.

Salinari, F., Giosue, S., Tubiello, F. N., Rettori, A., Rossi, V., Spanna, F., Rosenzweig, C., & Gullino, M. L. (2006). Downy mildew (*Plasmopara viticola*) epidemics on grapevine under climate change. *Global Change Biology, 12*, 1299–1307.

Salinari, F., Caffi, T., Legler, S.E. & Rossi, V. (2012). Implementation of epidemiological models for downy and powdery mildew in a DSS for integrated vineyard management. 1–5 October 2012: 10th EFPP (European Foundation for Plant Pathology) conference—"IPM2.0" towards future-proof crop protection in Europe, Wageningen (The Netherlands).

Seem, R. C., Magarey, P. A., McCloud, P. I., & Wachtel, M. F. (1985). A sampling procedure to detect grapevine downy mildew. *Phytopathology, 75*, 1252–1257.

Serra, S., Borgo, M., & Canotto, A. (1998). Primary infection occurrence in grapevine downy mildew. *IOBC/WPRS Bulletin, 21*(2), 5–7.

Schweizer, M. & Oliver, R.P. (1999). Molecular Fungal Biology. Cambridge University Press

Stryzik, S. (1983). Modèle d'état potentiel d'infection: Application a *Plasmopara viticola*. *Association de Coordination Technique Agricole*, Maison Nationale des Eleveurs, 1–46.

Tran Manh Sung, C., Strizyk, C., & Clerjeau, M. (1990). Simulation of the date of maturity of *Plasmopara viticola* oospores to predict the severity of primary infections in grapevine. *Plant Disease, 74*, 120–124.

Unger, S., Büche, C., Boso, S., & Kassemeyer, H.-H. (2007). The course of the colonization of two different Vitis genotypes by Plasmopara viticola indicates compatible and incompatible host–pathogen interactions. *Phytopathology, 97*, 780–786.

Vercesi, A., Toffolatti, S. L., Zocchi, G., Guglielmann, R., & Ironi, L. (2010). A new approach to modelling the dynamics of oospore germination in *Plasmopara viticola*. *European Journal of Plant Pathology, 128*(1), 113–126.

Vercesi, A., Tornaghi, R., Sant, S., Burruano, S., & Faoro, F. (1999). A cytological and ultrastructural study on the maturation and germination of oospores of *Plasmopara viticola* from overwintering vine leaves. *Mycological Research, 103*, 193–202.

Vercesi, A., Zerbetto, F., & Rho, G. (1999). Impiego dei modelli EPI e PRO nella difesa antiperonosporica del vigneto. *Frustula entomologica, 22*, 92–97.

Vercesi, A., & Liberati, D. (2001). Modelli epidemici: possibilità applicative e prospettive. *Informatore Fitopatologico, 51*(4), 13–18.

Viennot-Bourgin, G. (1949). *Les champignons parasites des plantes cultivées*. Paris: Masson.

Wong, F. P., Burr, H. N., & Wilcox, W. F. (2001). Heterotallism in *Plasmopara viticola*. *Plant Pathology, 50*, 427–432.

Yarwood, C. E. (1937). The relation of light to the diurnal cycle of sporulation of certain downy mildews. *Journal of Agricultural Research, 54*, 365–373.

Zachos, D. G. (1959). Recherches sur la biologie et l'épidémiologie du mildiou de la Vigne en Grèce. Bases de prévisions et d'avertissements. *Annales de l'Institut Phytopathologique Benaki, 2*, 193–355.